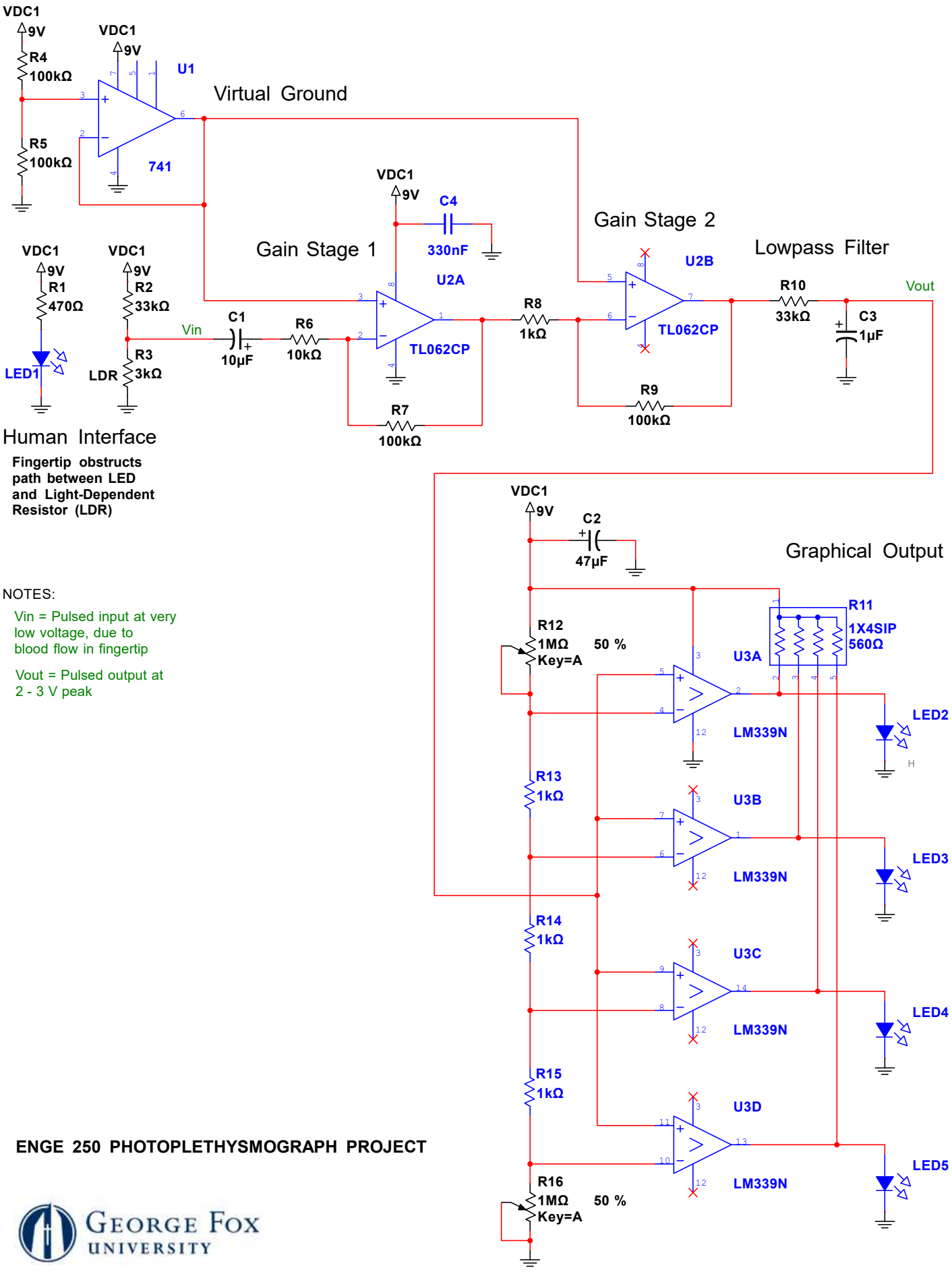

VDC1
9V
R4
100kΩ
R5
100kΩ
U1
741
Virtual Ground
Gain Stage 1
C4
330nF
U2A
TL062CP
Gain Stage 2
U2B
TL062CP
Lowpass Filter
R10
33kΩ
C3
1µF
Vout
R8
1kΩ
R9
100kΩ
R7
100kΩ
R1
470Ω
LED1
R2
33kΩ
R3
3kΩ
LDR
Vin
C1
10µF
R6
10kΩ
Human Interface
Fingertip obstructs path between LED and Light-Dependent Resistor (LDR)
C2
47µF
Graphical Output
R11
1X4SIP
560Ω
R12
1MΩ
50 %
Key=A
U3A
LM339N
LED2
R13
1kΩ
U3B
LM339N
LED3
R14
1kΩ
U3C
LM339N
LED4
R15
1kΩ
U3D
LM339N
LED5
R16
1MΩ
50 %
Key=A
NOTES:
Vin = Pulsed input at very low voltage, due to blood flow in fingertip
Vout = Pulsed output at 2 - 3 V peak
ENGE 250 PHOTOPLETHYSMOGRAPH PROJECT
GEORGE FOX UNIVERSITY